Xinchao Liu

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Education

University of Arkansas, Fayetteville
Industrial Engineering, Research Assistant 09/2019-Present
GPA: **4.0 out of 4.0**

Nanjing University of Aeronautics and Astronautics, Nanjing
M.Sc. in Aircraft Airworthiness Technology and Management 09/2016-04/2019

Nanjing University of Aeronautics and Astronautics, Nanjing
B.Eng. in Aircraft Design and Engineering (Helicopter Engineering) 09/2012-06/2016

Awards

Honorable Mention of College of Engineering 2022 Three-Minute Thesis Competition 1/2022
(*https://news.uark.edu/articles/58821/food-science-doctoral-student-takes-top-prize-in-three-minute-thesis-competition*)
Reginald R. "Barney" and Jameson A. Baxter Endowed Graduate Scholarship 1/2022
Winner of **INFORMS QSR** (Quality, Statistics and Reliability) **Best Student Poster** 10/2021
(*https://news.uark.edu/articles/58244/industrial-engineering-faculty-and-students-receive-awards-from-informs*)
(*https://connect.informs.org/qsr/awards*)
Travel grant from the University of Arkansas 9/2021
AAIE G. Kent Burnett Scholarship 5/2021
AAIE Larry Stephens Family Scholarship 5/2021
Second Place, the **poster competition** at Data Science Day (DSD) conference by the NextGen committee of New England Statistical Society 11/2020
(*https://news.uark.edu/articles/55381/industrial-engineering-student-receives-poster-competition-award-at-data-sciences-conference*)
Second Prize, the 14th China Post-Graduate Mathematical Contest in Modeling 12/2017
First Class Scholarship for Graduate 9/2016-3/2019
Excellent Student Scholarship 9/2012-6/2016
First Class Scholarship for freshman 9/2012

Graduate Coursework

Design of Industrial Experiment, Advanced Stochastic Process, Engineering Statistics, Multivariate Analysis, Finite Element Methods and solution of Sparse Linear Systems, Introduction to Optimization Theory, Engineering Application of Stochastic Process, Computational Statistics, System Simulation II

Conference Publications

- [C1] **Liu XC**, Liu X, Lin G. "Integration of physics and statistical learning: A Spatio-temporal Model for Nonlinear Structural Dynamics". ***INFORMS DMDA (Data Mining & Decision Analytics) Workshop***, 2020.
- [C2] Bao JP, Sun YC, **Liu XC**. "Research on Pilot Situation Awareness in Noise Environment". *4th International Conference on Electromechanical Control Technology and Transportation (**ICECT**)*, 2019.

Journal Publications

- [J1] Lu XH, **Liu XC**, Li YL, et al. "Simulation of airborne collision between a drone and an aircraft nose". ***Aerospace Science and Technology***, 2021, 107078. Impact factor: 5.107, ranking 2/34 in aerospace engineering.
- [J2] Lu XH, **Liu XC**, Zhang YC, et al. "Simulations of airborne collisions between drones and an aircraft windshield". ***Aerospace Science and Technology***, 2020, 98: 105713.
- [J3] Wang FY, Sun JZ, **Liu XC**, et al. "Aircraft auxiliary power unit performance assessment and remaining useful life evaluation for predictive maintenance". Proceedings of the Institution of Mechanical Engineers, Part A: ***Journal of Power and Energy***, 2019: 0957650919883718.

Written in Chinese language:
- [J4] Wang LC, **Liu XC**, Zuo HF. "Analysis of Bird Strike Resistance Effect of a Honeycomb Sandwich Structure of a Horizontal Tail". ***Aeronautical Computing Technique****, 2020, 50(2):37-40*.
- [J5] **Liu XC**, Xu YF, Wang LC, et al. "Parameter Study of Bird-strike Positions on Transport Airplane Windshield". ***Journal of Vibration and Shock***, 2019 (17): 14.
- [J6] Sun JZ, **Liu XC**, Liu RC, et al. "IDMS based method for quantitative monitoring of aero-engine ingested airborne sands". ***Acta Aeronautica at Astronautica Sinica***, 2017, 38(8): 177-187.
- [J7] **Liu XC**, Liu X, Guan TY, et al. "Rating and Inspection Interval of Environmental Deterioration (ED) and Accidental Damage (AD) for Civil Aircraft". ***Journal of Ordnance Equipment Engineering***, 2017, 38(8): 90-97.

Working Papers

- "Physics-Informed Statistical Learning for Nonlinear Structural Dynamics of Aircraft-UAV Collisions", submitted to *Technometrics*, under review, 2021.
- "DMD-VARX: Integration of Dynamic Mode Decomposition and Vector Autoregressive Model for Large-Scale Spatio-Temporal Dynamics with High-Dimensional Exogenous Variables", preparing, 2022.

Work Experience

- ***Structural strength engineer intern***, Shanghai Aircraft Airworthiness Certification Center, Civil Aviation Administration of China, Shanghai, China (July 2018 – October 2018).
  Main work: Finite element model validation and simulation runs on computer workstation
- ***Aircraft design and manufacture engineer intern***, AVIC Changhe Aircraft Industry (Group) Co., Ltd., Jiangxi, China (July 2013, July 2015).
  Main work: Manufacture process optimization

Invited Talks

- **Liu XC**, Liu X. "Statistical Learning for Nonlinear Structural Dynamics of Aircraft-UAV Collisions". *Symposium 316 – Dimension Model Reduction in Computational Mechanics and Engineering Systems*, ***EMI (Engineering Mechanics Institute Conference)***, 2022, Baltimore, Maryland.
- **Liu XC**, Liu X, Kaman T, Lin G. "Fast and reliable metamodeling of large-scale nonlinear time-dependent problems". *Session: Statistical Learning for Systems Analytics*, ***INFORMS***, 2021, Anaheim, CA.
- **Liu XC**, Liu X, Lin G. "Integration of physics and statistical learning: A Spatio-temporal Model for Nonlinear Structural Dynamics". ***INFORMS DMDA (Data Mining & Decision Analytics) Workshop***, 2020, virtual.
- **Liu XC**, Liu X, Lin G. "Integration of physics and statistical learning: A Spatio-temporal Model for Nonlinear Structural Dynamics". *Session: Advances in Statistical Modeling for Spatio-Temporal Data*, ***INFORMS***, 2020, virtual.

Current Research/Project Activities

Research area: Scientific Machine Learning, Uncertainty Quantification, Model Reduction, Digital Twin
Project: Data-driven Statistical Learning for the UAV-aircraft Mid-air Collisions; Physics-Informed Machine Learning for Additive Manufacturing.

Professional Affiliations

Student Member of SIAM (since 2020), IISE (since 2020), INFORMS (since 2019)
Member of UofA IISE Student Chapter (since 2019)
Member of UofA INFORMS Student Chapter (since 2019)
Member of Alpha Pi Mu Honor Society (since 2020)

Professional Service

Robotics service: **Referee & Robot Inspector**, 2021-2022 FIRST Tech Challenge (FTC) Arkansas Championship Tournament
Committee member of INFORMS QSR international session (2021-2022)
IISE Student **Leadership Board** (2021-2022)
Journal reviewer at the Proceedings of the Institution of Mechanical Engineers, Part G: ***Journal of Aerospace Engineering***. (2021)